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Chater 6 Electrical Maintenance

Summary

ZPMC integrated electrical control system is the big achievement resulting from ZPMC's pursue of innovation. This is not only breaking the foreign supplier's monopolization and lowering the cost, but also bringing reliability to the end user. The most important is to provide the end user the life-time technical support and software update service.

This STS crane maintenance manual also covers electrical part. The correct maintenance to the electrical system will assure the crane's availability and reliability. Correct and timely maintenance is also crucial to ensure the components' expected service life, such as motors, MV switchgear and transformers, etc.

This chapter will describe how to clean, maintain, check, trouble shooting and solve the problems of the key electrical components on the crane. It includes MV switchgears, MV transformers, motors, slip ring box, gear limit switches, load cells sensor and air conditions, etc. The maintenance stations, which are used, will be described here also.

It is necessary to point that, with the help of technical improvement, more and more computer technologies have been introduced to the ZPMC integrated electrical system. The trouble shooting is easier and easier, especially to the peripheral equipments. We can use indicator, fault screen to locate the problem directly. It is also easy to locate the cause of problem by searching PLC program. For detailed information of electrical system, please refer to the manual of ZPMC integrated electrical system.

As followed contents, just for reference, if you need the detail operation , please review the electrical manual.

6.1 Maintenance station

	**The maintenance station is purposely designed for maintenance reason. No other usage and non-related people are allowed.**

6.1.1 Over-travel bypass operation

Components and location	"over-travel bypass" key switch (on the PLC panel door）
function	1. Four main motions (hoist, trolley, boom and gantry) have its own over-travel limit switch. The control will be tripped, as soon as the limit switch is hit. The design of the limit switch will stop the control being put on again. 2. After checking and clearing all faults, maintenance person is needed to turn on the "over-travel bypass" key switch. Then, he should inform the crane operator to reset and put the control on. 3. The crane operator drives the relative motion (hoist, trolley, boom and gantry) to the opposite direction with a certain distance. 4. The crane operator stops all motions to let the maintenance people turn the "over-travel bypass" key switch back to original position. The crane is back to normal condition.



Figure 6-1：Overtravel bypass key switch

6.1.2 Snag load reset operation

Components and location	1. "Snag/Reset" pushbutton with red light (on the PLC panel door) 2. "Snag" red indicator (on right hand operator station) 3. "Snag bypass" key switch (on the PLC panel door)
function	1. In case of a snag accident, the control will be off and the "Snag" red indicator turns on. The driver should call maintenance people immediately. 2. Maintenance personnel should carefully check the crane. Make sure the crane has not been damaged and the snag cause has been solved. Then, the maintenance people can start to reset the snag condition. 3. Personnel should turn the "snag bypass" key switch on and inform the crane operator. The crane operator will put control on and lower the hoist a certain distance at slow speed. And stop the hoisting motion. 4. The maintenance personnel shall press the "Snag/Reset" pushbutton. This will cause the snag cylinder extended to normal position with the "snag" indicator flashing. When all cylinders extend to pre-set position, the "snag" indicator will be off. 5. After the maintenance people turn the "snag bypass" key switch to normal position, the crane is back to normal operation status.



Figure 6-2：Snag reset key switch



Figure 6-3：Snag bypass key switch

6.1.3 T/L/S local station

Components and location	1. “Manual-on/off” key switch (on local station) 2. “#1/2/3/4 cylinder extend/retract” select switch (on local station) 3. “Zero position setting on/off” key switch (on local station) 4. “Home” pushbutton with green indicator (on local station)
function	1. Turn on the “Manual-on/off” key switch to activate this station. 2. Zero position setting procedure. Active this local station by turn on the “manual-on/off” key switch. Control the “#1/2/3/4 cylinder extend/retract” select switch to adjust each cylinder position. After all cylinders move to zero position, turn on the “Zero position setting on/off” key switch. When all cylinders’ zero positions have been memorized by PLC, the “Home” indicator will be on. Turn off the “Zero position setting on/off” key switch. 3. Back to home position procedure. Active this local station by turning on the “manual-on/off” key switch. Press the “home” pushbutton. All cylinders will move to preset “home” position. Do not forget to turn “manual-on/off” key switch back to normal.



Figure 6-4：T/L/S local station

6.1.4 Emergency operator station / Rope reeving station

Components and location	1. “Emergency mode –normal/emergency” key select switch (on operator station) 2. “hoist /trolley/boom emergency select” 3-position select switch (on operator station) 3. “forward/up, stop, reverse/down” 3-positions select switch (on operator station)
function	1. Turn on the “Emergency mode-normal/emergency” key

	switch. 2. Select the working motion by using "hoist/trolley/boom emergency select" select switch. 3. Control the moving direction by using "forward/up, stop, reverse/down" select switch.

Emergency operation procedure:

1. Press the "e-stop" to ensure control will not be put on.
2. Position and fix the emergency motor to the right position.
3. Install the chain; Make sure the proximity switch will sense the chain correctly. If it is using fork type clutch to gear in the emergency motor, move the fork to connect the emergency motor on the mechanism. Make sure the limit switch has been activated.
4. Connecting the emergency motor power plug and turning on the power.
5. Turn on the "Emergency mode-normal/emergency" key switch. According the request, select the working motion by using "hoist/trolley/boom emergency select" select switch.
6. Toggle the "forward/up, stop, reverse/down" select switch to make sure all connection was doing correctly, and the brake is working fine.
7. Move the motions small distance and stop the emergency operation. Observe if the starting and stopping work fine or not.
8. If everything is fine, start the emergency operation.
9. After finishing the operation, turn off the emergency motor power. Pull out the power plug and reel the cable on the emergency motor.
10. Disconnect the chain. If it is fork type clutch, return the clutch to its original position.
11. Move the motor to its original position and fix it.

⚠	**During the emergency operation, arrange personnel to observe the moving mechanism, steel rope drum and A-frame (in boom motion). The communication between the observer and operator should be kept clean and reliable.**
⚠	**During emergency hoisting, the spreader cable reel drum will not be moving simultaneously. Personnel should be arranged to observe the condition of the spreader cable. Control the spreader cable winding up and down manually as per the instruction from the emergency motor operator.**

If the rope reeving device is also on the operator station, following is the description:

Components and location	1. "Rope reeving" key switch 2. "Rope reeving motor 1,2- forward/stop/reverse" select switch
function	1. Turn on the "Rope reeving" key switch when needed. 2. Operating the "Rope reeving motor 1,2- forward /stop/reverse" select switch to control the rope motor rotation.



Figure 6-5: Emergency operator station / Rope reeving station

6.1.5 Spreader cable reel operation

Components and location	1. "Manual-on/off" key switch (on local station) 2. "Winding on / Winding out" pushbutton
function	1. This station is used to manually adjust the spreader cable tension, when the spreader cable is over-tension or slackened. 2. Turn on the "manual-on/off" key switch. 3. According to the rope

	condition, manually control the "Winding on / Winding out" pushbutton. 4. Turn off the "manual-on/off" key switch after finished.



Figure 6-6: Spreader cable reel operate station

6.2 Medium Voltage switchgear

⚠	**Strictly following the relative electrical power operation safety rules. The MV switchgear is operated only by qualified personnel with guardian on site.**
⚠	**If there is grounding switch on the MV switchgear, close this switch before service.**
⚠	**Switch off the power before doing maintenance. Make sure nobody else can turn on the power**
⚠	**Please read carefully and following the maintenance manual coming with the equipment.**



Figure 6-7: MV switchgear

Figure **6-7**: Shows the MV switchgear used in the machinery house on the STS crane. The maintenance of this MV switchgear will be as following:

6.2.1 Daily inspection

The daily inspection is to identify the fault as early as possible. The inspection includes at least following contents:

- Check if the cabinet gates are close properly or not;
- Check if the heater works fine or not;
- Check if there is any abnormal smell and/or noise or not;
- Check if the instrument works properly or not;
- Check if the relays works fine or not;
- Check if the power-on indicator works fine or not;

6.2.2 Periodic inspection

The periodic inspection is to keep the MV switchgear works trouble free and prolong its service life. The service period is depending on the environment, operating frequency and service years. Following is the suggestion for the service period.

- Every 2~3 years, if the environment is good and infrequently operating the MV switchgear;
- Every year, if the MV switchgear operation is more than 10 times per month;
- Every half year, if the environment is bad and the MV switchgear is operating frequently;
- If the circuit breaker is at spare state, doing the visual inspection every half year is recommended.

6.2.3 Periodic inspection items

The periodic inspection includes following items:

- Check if there is damp, rust or dirty inside the cabinets, on components or not;
- Check if the bolt tightening torque is enough or not;
- Check the function of switching components, controlling, interlocks, protection, signaling and other equipments;
- Visual inspection of contactor for the sign of discharge, burn or abrasion. If there is above mentioned sign, suggest replacing the contactor;
- Check the switch accessories, peripheral equipments and the insulation board. They should be kept dry and clean.

6.2.4 Items to be tested

While doing the periodic inspection, the following performance test is suggested to be done at the same time:

- Operating the switch 5 times to check if the control circuit is normal or not;
- Check if the interlocks between panel works fine or not;
- Electrical test, mainly the insulation test.

6.2.5 MV switchgear cleaning

When there is dirty on the switchgear, it should be cleaned carefully, especially on the insulating material.

- For the floating dust, we suggest to clean it with dry cloth;
- For emplastic / lipidic type dirt, we need use dry cloth with family-used light alkalescent cleanser. After that, clean it with wet cloth. Finally dry it;
- For the insulating material or heavily polluted component, halogen free cleanser should be used.

6.2.6 Out of service items

If following situation found during inspection, the MV switchgear should be taken out of service and repair:

- Bad insulation, discharge, flashover or strike;
- Damaged components;
- Other abnormal phenomena which affect safety operation;

6.2.7 Safety list for MV switchgear service

Before starting, the service people should check item by item according the "Safety list for MV switchgear service" (so called "LIST"). If the service has been stopped, before restarting, the service people should check the "LIST" again.

- Before starting, the service people should understand the site arrangement, including power-on or power-off areas, grounding wire arrangement, control power status, emergency power availability, on-site guardian availability and safety ways, etc.
- Check whether the safety measures required by safety rules have been ready or not. The service people should check it personally regarding the safety measures, isolation measures between on-duty products and service products, isolation signs, etc.
- Before servicing, it is better to turn off all power feeding to this equipment. For any equipment with "star" type connection, the neutral point should be disconnected also. It is forbidden to working on an equipment with only circuit breaker opened, the disconnect switch should be opened to "test" or "service" position also. Please be noticed, it should always disconnect both side of transformer and PT.
- Be cautious of the cabinet number. Always avoid working on a wrong cabinet.
- Only one grounding point is permitted at the second side of CT and PT.
- When entering the cabinet or moving the equipment out for service, be cautious not to hurt anybody.

6.3 Transformer

	**Strictly following the relative electrical power operation safety rules. The transformer is operated only by qualified personnel with guardian on site.**
	**Before any maintenance, fully discharge both MV side and LV side. Separate the transformer from the incoming and outgoing. Grounding all transformer terminals.**
	**If there is a grounding switch on the MV switchgear, turn on this switch before doing any maintenance on transformer.**
	**Switch off the power before doing maintenance. Make sure nobody else can turn on the power.**
	**Do not power on the transformer if there is any condensate water on the transformer.**



Figure 6-8: Transformer

6.3.1 Maintenance period

In order to keep transformer running in good status and prolong its service lift, it is important to do the transformer maintenance periodically. The maintenance time is depending on the external factors, such as the environment, etc.

- First maintenance service can be done one month after transformer is energized.
- In dry and clean environment, the maintenance can be done every year.
- In dusty and chemical fog environment, the maintenance can be done every 3-6 months.
- In heavily polluted environment, the maintenance should be done every month.

6.3.2 Period inspection item:

- Check every fastening device of the transformer. Fasten it when needed.
- Check all connection points in the power-circuit, any loose will cause overheating on the connection. Check if there is any stain, rust on the electric and other parts or not.
- Check if accessories' circuits are normal or not.
- Check the grounding of transformer itself and covers.
- Check the transformer winding resin for cracks, electric corrosion or carbonization. If any damage is found, please contact the transformers' manufacturer.
- Measure the insulation resistance and record the value.

6.3.3 Transformer cleaning

Periodic cleaning, especially the surface cleaning, is very important to maintain the transformer in good cooling and insulation level. The cleaning should be done as following:

- Do a comprehensive visual inspection for coils, leads and the temperature monitor.
- Check if there is any foreign substance inside or outside of the transformer. Remove it if anything found.
- Clean the insulators, leads, insulating spacers and windings. The dust removal can be done by vacuum cleaner or air compressed with dry air/nitrogen. The compressed air/nitrogen should be no more than 0.2 MPA. The surface of the insulators, wires, terminal plates and other insulated parts should be cleaned by dry cloth.
- Do the maintenance on accessory device according the relative manual.

6.3.4 Preventive test

The preventive test is suggested to be done every three years. It should be done immediately if any fault happens. The test includes voltage withstands, transformer ratio error check and percent impedance etc.

6.3.5 Bolt torque verification

One month after transformer is in operation, check the fastening torque of every bolt according the table **6-1**:

table 6-1：Bolt torque check table

Bolt size (mm)	M8	M10	M12
Torque (N.m)	21	30	50

6.3.6 Normal faults and solutions

The common faults and solutions for the transformer are as per:

Table 6-2：Normal faults and solutions

Faults	Possible Cause	Solutions
Insulation too low	Winding is wet or with dust on the surface	Remove the dust with dry air
	Aging insulation	Contact the manufacturer
MV side voltage abnormal	MV side voltage too high or too low	Check installation and contact the MV power supplier
LV side voltage abnormal	MV side voltage too high or too low	Check installation and contact the MV power supplier
	Wrong tapping connection on MV side	Change tapping connection
	Loose tapping connection on MV side	Tighten tapping connection
	Winding not continuous	Contact the manufacturer
	PT or voltmeter abnormal	Replace with a new one
Three phase unbalance	Wrong tapping connection on one phase	Change tapping connection
	One of the fuse on MV side damaged	Replace with a new one
	Load unbalance	Change load arrangement and balance it
	Connecting bolts loose	Check installation and tighten
	PT or voltmeter abnormal	Replace with a new one
Upper steam protection tripped when working or putting in operation	Thermal protection setting incorrect	Change setting
	Thermal protection malfunction	Replace it with a new one
	Temperature sensor malfunction	Replace it with a new one
	Wrong type of fuse or protective relay	Check the actuation time and current
	Overload	Check the load
	Winding insulation damaged	Contact the manufacturer
	Wrong CT wiring	Check the polarity and connections
Insulation damaged or Smoky	Continuous overload	Contact the manufacturer
	Dust or fouling on winding	Contact the manufacturer
	Mechanical damage during transportation or installation	Contact the manufacturer

Faults	Possible Cause	Solutions
	Lighting rod or surge absorber damaged	Contact the manufacturer
Temperature abnormal	Continuous overload	Contact the manufacturer
	Bad Ventilation	Improve ventilation
	Environmental temperature too high	Cooling by air-condition or fan
	Iron core abnormal overheating caused by under frequency on MV side over voltage	Check the MV side connection
Vibration and noise	MV side under frequency or over voltage	Check the MV side connection
	Framework loose during the transportation or installation.	Tighten it
	Reflection voice caused by too close to the wall or other things	Remove obstacle
Transformer under efficiency	Overload	Check the load
	Under frequency	Check the frequency
	MV side over voltage	Check MV side voltage
	Wrong tapping connection	Check tap connection
Terminal overheating	Wrong bolt connection	Tighten it

6.4 Three-phase AC motor

	**Before any maintenance, make sure all motor circuits have been turned off. Don't forget turn off the accessories' circuits, such as heaters.**
	**Some parts will be high temperature after running.**
	**Switch off the power before doing maintenance. Make sure nobody else can turn on the power.**



Figure 6-9：Three-phase AC motor

Figure **6-9：** shows the three-phase AC motor under in machinery house for boom motion. The maintenance items will be as following:

6.4.1 Motor inspection period and items

1) If the crane is out of operation for long time, the motor should be run at least once per month. The motor heater should be kept working.

2) If the motor is just repaired, or has been put into service for 500 hours or 1 year (whichever comes first), the following inspection should be done:

- When motor is running, check if the electrical and mechanical characteristics are qualified or not. For example, check if there is any excessive current.
- When motor is running, check the smell, vibration, and noise. Burning smell can be caused by overheated winding; while vibrations and noises can be caused by some mechanical faults. If any burning smell or abnormal vibration or noise found, please turn off the motor immediately.
- When motor is running, check the bearing temperature and oil leakage.
- When motor is not running, check if there is any crack on the motor or not.
- When motor is not running, check the motor inside. Motor internals should be away from water dripping, oil and foreign substances. Air inlet and outlet should be unobstructed. Drain holes, if exist, should be unblocked.

3) If the motor is putting into service for 1600 hours or 2 years (whichever comes first), the following inspection should be done:

- When motor is running, check if the electrical and mechanical characteristics are qualified or not. For example, check if there is any excessive current.
- When motor is running, check the smell, vibration, and noise. Burning smell can be caused by overheated winding while vibrations and noises can be caused by some mechanical faults. If any burning smell or abnormal vibration or noise found, please turn off the motor immediately.
- When motor is running, check the bearing temperature and oil leakage.
- When motor is not running, check if there is any crack on the motor.
- When motor is not running, check the motor inside. Motor internals should be away from water drippings, oil and foreign substances. Air inlet and

outlet should be unobstructed. Drain holes, if exist, should be unblocked.

- When motor is not running, check if the motor installation is still OK or not.
- When motor is not running, check coupling and installation bolts. Make sure there is no discoloration of the cables and insulation parts.
- When motor is not running, check if the winding insulation is good enough or not.
- When motor is not running, make sure the insulated bearing, if adopted, is not bypassed.

6.4.2 Motor lubrication and lubrication grease

- The type of lubricating grease and the lubricating interval should follow the motor manual or nameplate. Normally, the minimum temperature for general lubricating grease operable is -20° C.
- The following lubricating interval is only applicable for horizontal installed motor whose working temperature is less than 40° C. This lubricating interval should be shortened when it is variable frequency motor, or vertical installed motor, or the high speed running motor, or motor mounted in vibrating place, with impact load, or frequently forward/backward running motor.
- Different brands of lubricating grease should not be mixed, the amount of lubricating grease should follow the motor manual requirement, Never over-lubricate the motor.
- When it is permanent greased 4-pole motors, the lubricating interval is 20000-40000 hours. When it is permanent 2-pole or 2/4 poles motors, the lubricating interval is 10000-20000 hours. The above lubricating intervals should be shortened as for the big motor.
- When it has grease nipple for the motor bearing, the maximum lubricating interval is 12 months.

6.4.3 Motor filter cleaning

The motor filter needs regularly cleaning and replacement. Excessive dust on the filter will block the air getting into the motor and cause the motor overheating. The filter can be cleaned as the following:

- Loose the fastening bolt and take out the filter together with its frame. When cleaning the filter, never crumple or twist it.
- Clean the filter by warm water.
- If there is oil on the filter, dip the filter into cleanser for about 10 minutes before cleaning.
- After the filter is dry, clean it by compressed air.
- Clean the floating dust on the filter by vacuum cleaners.
- Install the filter back in right direction.
- If any hole or fiber broken found, the filter needs to be replaced.

6.4.4 Normal faults and solutions

The normal faults and solution of motors as per **Table 6-3**:

Table 6-3：Normal faults and solution

Faults	Possible Cause	Solutions
Unable to start	Stator phase open	Check the stator winding, fix the open phase.
	Short circuit between circuits or phases on the stator	Check if the resistance and the non-load current of each phase is balance or not. Find the short circuit and fix it.
	Wrong connection on stator	Correct the stator winding wiring according to the wiring diagram on the nameplate.
	Overload or gearing mechanical fault	Separate the motor from the load, if the motor is able to start, check the gearing.
	Drive parameter setting wrong	Check the drive parameter and make adjustment.
Speed lower than rated	Wrong parameter setting of output voltage and frequency	Reset the parameter according to the requirements
	Load too heavy	Check the gearing
Abnormal noise or vibration	Mechanical friction (between stator and rotor)	Check the air gap between stator and rotor, find out the reason and fix it.
	Phase lose running	Run the motor again, if it is unable to restart, one phase may lose, check the power supply and the motor to fix it.
	Bearing lack of grease or damaged	Clean the bearing and lubricate it, or replace with a new one.
	Motor wiring wrong	Find out and fix it
	Rotor unbalanced after repair	Make it balanced
	Motor shaft distorted	Make it straight or replace with a new one if needed
	Coupling loose	Find the loose bolt and tighten
	Installation base not horizontal or with flaw	Check the installation base and make it horizontal

Faults	Possible Cause	Solutions
Motor overheating	Over load	Check the current on drive panel display or measure the stator current by electromagnetic ammeter. If overload found, please reduce the load.
	Phase-lose running	Check the wiring of the drive and the motor. Fix it when needed.
	Motor connection wrong	Mistake Y connection for △ connection, or mistake △ connection for Y connection. Turn off the power and change the connection ASAP.
	Stator winding short-circuit	Find the grounding or short-circuit part and fix it.
	Squirrel-cage rotor broken	Replace it with a new one, if needed
	Friction between stator and rotor	Check if the bearing, stator and rotor are in good installation level or not. Fix if it is not good.
	Poor ventilation	Check if the fan is damaged, or the air duct is blocked. Repair the fan if damaged. Remove the obstructions and clean the duct to make good ventilation.
	Motor over-excitation when running in low speed caused by wrong U/f parameter setting	Adjust the U/f parameter setting
	Braking current too high when using inverter's DC braking function	Adjust the braking current setting according to the braking frequency. Generally the setting is 100%-150% of rated current.
Bearing over heating	Bearing damaged	Replace with a new one
	Bearing grease too much or too less or impure.	Adjust or replace the grease
	Assembly between bearing and axis, bearing and cover too loose or too tight	Fit it
	Poor installation of motor or bearing cover	Make it fit and tighten the bolt
	Poor installation of the oil seal for IP54 motor	Adjust the installation
	Poor installation of belt or coupling	Adjust the installation
Motor casing electrified	Bad grounding	Check the grounding of the motor casing
	Winding damped or insulation too low	Dry it
	Stator winding insulation too low	Repair the motor
	Dirty terminal board	Clean it.
	Out-lead insulating wears out	Wrap it with insulation material.

6.5 Special cable

Some special cables need periodic inspection which is basically a visual inspection. One of the inspection methods is to make a marking line on the cable surface. (If the marking line is straight, this means the cable is in good condition. If the marking line is twisted, attention needs to be paid). The other method is to observe the cable surface condition (such as cracks). Table 6-4 gives some examples.

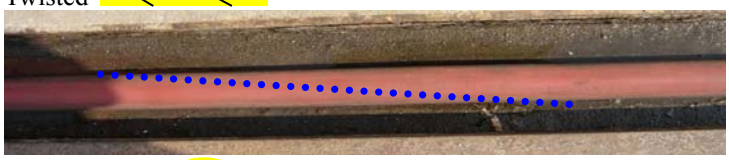
Under normal condition, the MV reeling cable should be replaced every 10 years. The spreader cable should be replaced every 5 years or every 750,000 TEU. The festoon cable can be replaced every 5 years or every 750,000 TEU. After the service life, If no serious distortions are found, these cables may be used as extended time as per the actual condition.

Table 6-4：Cable deformation examples

Cable	Type	Appearance	Solution
Cable for spreader cable reel	CORDAFLEX	Twisted cable	
		Expanded cable	Contact the supplier
Cable for spreader basket	SPREADERFLEX	Spiral cable	
Trolley power cable	CHAINFLEX	Cracked cable	
HV cable	PROTOLON	Crinkly cable	
		Spiral cable	
		Cracked cable	

Attached, please find the picture of these examples

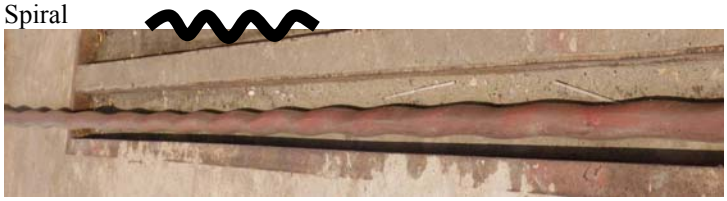
Twisted



Expanded



Spiral



Cracked



6.6 Air conditioner

⚠	**Before any maintenance, make sure all air-condition's power circuits have been turned off.**
⚠	**Periodic maintenance is very important to the air conditions' performance and life.**
⚠	**Switch off the power before doing maintenance. Make sure nobody else can turn on the power.**



Figure 6-10：Air conditioner

Figure **6-10**： shows the air conditioner. Maintenance of the crane air conditioner can be done as follows:

6.6.1 Monthly maintenance items

- Clean air-conditioner's filter

Air-condition filter is usually installed at return air inlet according to the environment. It can effectively filter dust, pollen and other impurities, etc in the air in order to protect the evaporator.

Air-condition filter is usually made of textile fiber or leno. If it is not cleaned timely, the air volume will get lower and the refrigerating effect gets worse.

To clean the filter, the service people can put it on a hard even place, and flap it gently to remove the big impurities. Wash it by warm water with neutral cleanser if needed. Install it back until dried.

If any hole or fiber broken found, the filter needs to be replaced.

- Check rubber belt (only for indoor unit with belt driven)

If the rubber belt is under tension, the friction between the belt and the belt pulley will be too small to drive fan blade, which will cause the air volume getting lower and the refrigerating effect getting worse. However, if the rubber belt is over tension, the bearing and axis of the fan motor will become easy to wear out, which will reduces the fan's service life. Either the rubber belt too loose or too tight will be harmful to the fan system and increase noise.

One week after air-conditioner is in operation, readjust the rubber tension properly. Later on, please do periodic inspection every month. Adjust the rubber belt or replace it if needed.

- Condenser cleaning

Condenser is the heat exchanger of air condition to exchange heat to outdoor. Usually it is installed outdoor, so it is easy to get dirty. If the condenser is not cleaned timely, it will affect the heat disperse level of air condition, which will cause the condenser over pressure and the refrigerating effect getting worse.

The condenser fin is very sharp, so any incorrect operation could result in injury. Please be very careful when cleaning the condenser. Usually, it can be cleaned by vacuum cleaner together with nylon brush. If compressed air available, the dust on the surface can be cleaned by the compressed air.

- Condensate system inspection

For air-conditioner on cranes, it is important to do periodic inspection of condensate system. Any blockage or leakage will cause serious consequence.

Clean the condensed water system periodically. Check if there is any leakage or blockage of discharge pipe. Check if the heat preservation is in good level or not.

- Compressor system inspection (suggested to be done by qualified air condition technician)

Compressor system is the heart of refrigerating system. Any small fault will directly affect the refrigerating capacity.

Check if suction pressure and discharge pressure are normal by watching the charging meter. Check if there is any oil leak in the system.

6.6.2 Yearly maintenance items

Besides the monthly maintenance, the following maintenance needs to be done every year:

- Check if there is any dust or mildew on the surface of evaporator. Clean it if needed.
- Check if the bearing of the fan motor wore out.
- Check if the installation frame of the outdoor unit is rusted.
- Check if there is any loose connection of the electric wiring.

6.6.3 Normal fault and solution

Table 6-5：Normal fault and solution

Faults	Possible Cause	Solutions
High pressure protection	Air in refrigerating system	Draw vacuum and charge refrigerant
	Refrigerant too much	Discharge excess refrigerant
	Condenser too dirty	Clean it
	Condensate air inadequate	When refrigerating, check the outdoor unit. When heating, check the indoor unit.
	Condensate pipe blocked	Check the flow capacity of drier-filter and expansion valve
	High pressure switch fault	Replace with a new one
Discharge temperature protection	Condensing pressure too high	Check and fix
	Suction superheat too high	Refrigerant not enough, or with leakage or blockage
	Motor overheating	Find the reason of poor lubrication and fix it
Discharge superheat too low	Suction with liquid	Check and fix
Low pressure protection	Refrigerant insufficient or leak	Fix the leakage and charge refrigerant
	Evaporator too dirty	Clean it
	Evaporator with serious frost	Check if the defroster is working
	Evaporation air inadequate	When refrigerating, check the indoor unit. When heating, check the outdoor unit.
	Suction pipe blocked	Check and fix
	Condensate pipe blocked	Check the flow capacity of drier-filter and expansion valve
	Low pressure switch fault	Replace with a new one
Compressor abnormal noise	Poor lubrication	Check and fix
Circuit breaker tripped	Short circuit	Check if there is any short circuit of electrical wiring, or motor winding burnt
Compressor overheating	Discharge pressure too high	Check and fix
	Suction air inadequate	Check and fix
	Poor lubrication	Check and fix
	Crankcase jammed	Replace the compressor
Temperature sensor fault protection	Temperature sensor fault	Replace with a new one
Refrigerating /heating inadequate	Refrigerant insufficient or leak	Fix the leakage and charge refrigerant
	Air filter too dirty	Clean it.
	Heat exchanger too dirty	Clean it
	Belt slipping	Adjust the belt, replace it if needed.
	Compressor fault	Check the cause of insufficient discharge and fix
	Thermostatic expansion valve fault	Adjust it or replace it if needed
	Fan direction wrong	Change its direction
	Temperature sensor faulty	Replace with a new one
Unit not working	Power off or fuse broken	Check the power, change the fuse if needed
	Wrong connection	Correct it according to the drawing
	Under voltage	Check it
	Phase reversal	Correct it
	Last fault not reset yet	Reset it
Indoor unit leakage	Insulating layer broken	Fix it
	Indoor unit not horizontal installed	Adjust its installation position
	Drainage blocked	Clean the drip tray and the drain pipe
	Drainage loose connection	Fix it and seal it
	Drain pipe installed wrong way	Re-design the drainage pipe
Abnormal noise when working	Foreign substance inside unit or motor	Remove it
	Poor installation of unit, or loose bolts	Adjust the installation, fix the bolts

6.7 Low voltage cabinet

⚠	**The LV cabinet covers electric cabinet, operation station, module box and junction box, etc. In order to keep crane's normal operation, maintenance of LV cabinet is needed to be done. When doing the maintenance, please always following the safety regulations/rules/requirements issued by the power supply department or the customer's company.**
⚠	**Switch off the power before doing maintenance. Make sure nobody else can turn on the power.**



Figure 6-11：LV cabinet

Figure **6-11**： shows the LV cabinet in the electrical room. The maintenance of LV cabinet related to the following items:

6.7.1 Low voltage cabinet period maintenance

Check if the reading of the meter on panel door meets the requirement of the designed capacity or not; check if indicator lights and operation stations are normal or not; check if there is any abnormal noise, smell or temperature or not; and check if there is serious dust deposit on the filter or not.

Recommend to do comprehensive maintenance every year. According the environment and the maintenance period, the maintenance can be done as following:

- Remove dust by brush, vacuum cleaner or dry cloth. Be careful not to blow the dust into any electric component.
- Check connected screws, wiring connections, operation stations, interlock switches. Replace it if any problem found.
- Check if the insulation resistance is OK for operation. If not, check and fix it immediately.
- Replace the arc chute and insulating parts whichever broken is found.
- Check grounding.

Periodically inspect the operation station function, especially the seldom-used function. The inspection can be done as following:

- Emergency stop. After control on the crane, push the emergency stop on different operation stations. Make sure the emergency stop on every operation station is working and will trip the control as soon as it is pushed.
- Lamp test pushbutton. When lamp test pushbutton is pressed, all indicator lights should be on, and buzzer buzzing.

6.7.2 Low voltage cabinet inspection period

table 6-6 describe the inspection period of low voltage cabinet. "√" means inspection needed. "-" means no need.

table 6-6： Low voltage cabinet inspection period

Items	Components	Inspections	Time interval					Solutions if problem found
			Every month	Every 3 months	Every year	When running	When stop	
Wiring	Entire crane	Electric connection	-	-	√	-	√	Tighten
	Headblock and spreader	Electric connection	√	-	-	-	√	Tighten
	Trolley	Electric connection	-	√	-	-	√	Tighten
equipments	Switches	Function	-	√	-	-	√	Replace
		Cover outlook	-	√	-	-	√	
	Circuit protection	Function	-	√	-	-	√	
		Cover outlook	-	√	-	-	√	
	Drive fan	abnormal noise or vibration	-	√	-	√	-	Replace the fan
		20000 hours'	-	√	-	√	-	

Items	Components	Inspections	Time interval					Solutions if problem found
			Every month	Every 3 months	Every year	When running	When stop	
		operation						
	Cabinet fan	abnormal noise or vibration	-	√	-	√	-	
		20000 hours' operation	-	√	-	√	-	
Grounding	Horizontal and vertical installed copper bar	Over temperature	-	√	-	√	-	Check
		Poor insulation	-	√	-	-	√	Replace
		Loose bolt	-	-	√	-	√	Tighten

6.8 Slip ring Box

	**Strictly following the relative electrical power operation safety rules. The MV cable reel slip ring box is operated only by qualified people with guardian on site.**
	**Before any maintenance of spreader cable reel slip ring box, make sure all the power circuits to the spreader cable reel slipping box have been turned off. Especially the accessories' circuits, such as heaters.**
	**Switch off the power before doing maintenance. Make sure nobody else can turn on the power.**



Figure 6-12: Gantry cable reel slip ring box

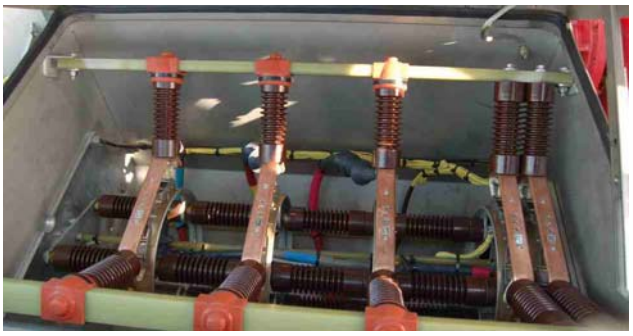


Figure 6-13: Gantry cable reel slip ring box inside

Figure **6-12:** and Figure **6-13:** shows the gantry cable reel slip ring box. It is located on one side of Sillbeam; near the gantry cable reel. The spreader cable reel slip ring box is on the trolley. The maintenance of slip ring box includes the following items:

6.8.1 Slip ring box check item

- The surface of slip ring should be kept clean, without any dust or oil fouling. In the environment where the slip ring will easily get dirty, periodic inspection for the surface of slip ring and carbon brush is needed.
- If any spot found on surface of slip ring, please polish it with sand paper.
- Check the carbon brush frequently to insure good contact between carbon brushes and slip rings.
- Check the tightening torque of every bolt in slipping box.

6.9 Gear limit switch

Sometimes, the gear limit switch needs to be adjusted, such as changing rope, steel rope stretch after using, loose coupling of Gear limit switch, etc. Attached, please find the description for gear limit switch adjustment.

1 Hoist Gear Limit Switch adjustment:

Items	Description
Change steel rope	1. Before changing the steel rope, put the spreader down to the ground and leave the steel rope tensioned. 2. Uncouple the gear limit switch. Be careful not to let it turned. Keep the gear limit switch away form other people. Mark a line from the shaft to its cover. 3. Bypass the gear limit switch signals in the PLC, and finish the rope changing. Bypass the hardware circuit also if needed. 4. Put the spreader down to the ground, inch the hoist to the same position as made in "Step1". Re-couple the gear limit switch to the coupling. 5. After that, cancel all bypasses either in software or hardware. 6. Operate the hoist motion up and down slowly to verify the installation of the gear limit switch. If all gear limit switch positions are obviously different from the original, first check if the coupling and the steel ropes are OK or not, then uncouple the gear limit switch and adjust the shaft. If there is only certain gear limit switches out of position, adjust these gears separately. 7. Operate the hoist in slow speed and make sure the gear limit switch is OK; gradually speed up until full speed. Reconfirm the gear limit switch. 8. After confirming the gear limit switch is working OK; inform the operator that the crane can be back to service.
Steel rope being stretched	1. Steel rope will be stretched longer after certain times of using. All gear limit switch positions will be shifted downwards. During the operation, if no error reported because of the gear limit switch position, do not adjust the gear limit switch too often. 2. If all positions of gear limit switch are all wrong. Before uncoupling the gear limit switch, please observe the offset that need to be adjusted and make corresponding record. 3. Uncouple the gear limit switch and make corresponding adjustment according to the record, and couple it. 4. Let the operator hoist up and down, and make sure all gear limit switches are in right position. 5. Repeat the above procedure if needed.
Gear limit switch coupling loose	Refer to item "Steel rope being stretched"

2 Boom Can limit switch adjustment

Items	Description
Change steel rope	Refer to the corresponding item in hoist gear limit switch adjustment. Please pay attention that when marking on the gear limit switch, keep the boom horizontal and the steel rope tensioned.
Steel rope being stretched	Refer to the corresponding item in hoist gear limit switch adjustment. The position to judge weather the boom steel rope is too loose or not should be boom level position.
Gear limit switch coupling loose	Refer to the corresponding item in hoist gear limit switch adjustment.



Figure 6-14: Gear limit switch

6.10 Load cell sensor

Load cells convert load weight to electrical signal (usually 4~20mA signal). After the electric signal goes into the PLC analog-input module, PLC calculates this analog-input value to the load weight. If no special

requirement, generally the load cell sensor outputs 4mA when spreader on the ground and steel rope slack, 20mA when rated load. Please find the following diagram**Figure 6-15：**。Verify the load cells every year. Check the load cell linearity, zero load, and full load condition with three load conditions (empty spreader, half load and rated load). Compare the actual weight with the PLC value. If the difference is too big, please make corresponding adjustment in PLC.

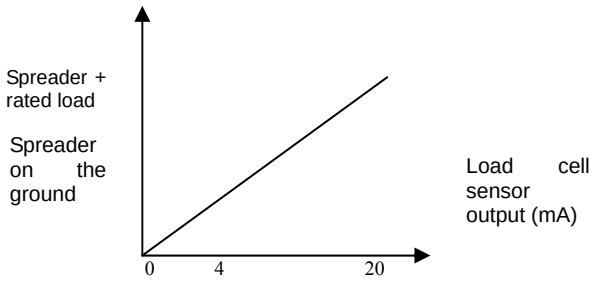


Figure 6-15：Curve of load cell current vs load

Table 6-7：Normal faults and solutions

Faults	Possible Cause	Solutions
Output current lower than 4mA	Wrong setting of load cell sensor	Return to the manufacturer to reset
	Wrong installation of the load cell sensor	Check the force direction and adjust it
Output current 0mA	Cable damaged	Check the cable
	Internal amplifier or strain gauge damaged	Return to the manufacturer
Output current higher than 20mA	Wrong setting of load cell sensor	Return to the manufacturer to reset
	Overload	Reduce the load
Output current higher than 20mA or 25mA	Wrong setting of load cell sensor	Return to the manufacturer to reset
	Overload	Reduce the load
	Short-circuit of the analog feedback circuit	Check the wiring and connection
	Internal amplifier damaged	Return to the manufacturer



Figure 6-16：Load cell sensor

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